

GRODNO, Belarus

WMO: 268250

Lat: 53.602N

Lon: 24.056E

Elev: 149

StdP: 99.55

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.1	-15.7	-22.0	0.5	-18.9	-18.3	0.8	-15.4	12.5	4.2	11.2	2.0	1.9	350	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	29.4	20.0	27.3	19.1	25.7	18.1	21.1	26.9	20.1	25.7	19.1	24.1	3.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.2	23.9	18.1	13.3	22.5	17.2	12.5	21.4	61.7	27.1	58.3	25.5	55.1	24.3	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.1	8.8	7.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-22.8	32.3	3.8	2.1	-25.5	33.8	-27.8	35.0	-29.9	36.2	-32.6	37.7
			-23.0	22.9	3.7	1.3	-25.7	23.8	-27.9	24.6	-30.0	25.3	-32.7	26.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-3.7	-2.5	1.5	8.2	13.2	16.1	18.9	18.1	13.1	7.3	2.6	-1.7		
	DBStd	9.03	5.96	5.60	4.49	4.35	4.03	3.23	3.19	3.23	3.32	3.97	4.32	5.45		
	HDD10.0	1833	424	349	263	85	14	1	0	0	11	101	223	363		
	HDD18.3	4009	682	583	521	306	166	82	31	44	160	341	473	621		
	CDD10.0	977	0	0	1	30	113	184	277	252	103	18	1	0		
	CDD18.3	111	0	0	0	1	6	16	50	37	2	0	0	0		
	CDH23.3	1004	0	0	0	8	68	137	448	322	22	0	0	0		
	CDH26.7	252	0	0	0	1	8	21	131	89	3	0	0	0		
(14) Wind	WSAvg	3.5	4.1	4.0	3.9	3.6	3.3	3.0	2.9	2.9	3.2	3.4	3.8	4.0		
(15) Precipitation	PrecAvg	582	36	33	33	36	58	63	86	59	52	44	43	40		
	PrecMax	700	79	61	73	76	115	125	173	162	136	128	75	66		
	PrecMin	443	10	10	10	9	23	18	5	5	14	2	12	16		
	PrecStd	67	15	12	15	19	21	28	44	33	35	30	15	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	8.7	15.1	24.3	27.8	29.0	32.7	32.7	25.9	19.5	12.7	8.9		
		MCWB	6.6	6.2	9.3	14.6	18.0	20.0	20.5	20.3	16.9	15.1	10.5	7.6		
	2%	DB	5.6	6.8	12.2	20.8	25.2	26.8	30.1	29.0	22.9	16.6	10.8	7.1		
		MCWB	4.6	5.3	7.9	12.9	17.3	18.6	20.0	20.3	16.1	13.4	9.5	6.2		
	5%	DB	4.1	5.1	9.6	18.1	23.0	24.9	27.9	26.8	20.8	14.6	9.1	5.7		
		MCWB	3.2	3.9	6.4	11.1	15.8	17.8	19.6	18.7	15.4	12.2	8.1	4.9		
	10%	DB	2.8	3.8	7.3	15.7	20.8	22.9	25.9	24.8	18.9	13.0	8.0	4.1		
		MCWB	2.2	2.8	5.1	9.9	14.7	16.7	18.6	17.6	14.3	11.2	7.1	3.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	6.9	10.4	16.0	19.7	21.5	23.0	22.4	18.6	15.9	11.2	7.9		
		MCDB	7.4	8.0	13.8	23.8	25.6	27.4	29.6	28.2	23.7	19.0	12.5	8.5		
	2%	WB	4.7	5.3	8.2	13.4	18.1	19.9	21.4	20.9	17.0	14.1	9.7	6.3		
		MCDB	5.3	6.3	11.3	19.2	23.6	25.0	27.0	27.0	21.1	15.9	10.6	6.8		
	5%	WB	3.2	4.0	6.6	12.0	16.8	18.7	20.4	19.7	15.9	12.6	8.2	4.9		
		MCDB	3.8	4.9	9.0	17.1	21.7	23.4	25.9	25.3	19.3	14.1	8.9	5.6		
	10%	WB	2.1	2.8	5.3	10.6	15.3	17.4	19.4	18.6	14.9	11.4	7.2	3.5		
		MCDB	2.7	3.7	7.1	14.5	19.7	21.5	24.5	23.1	18.1	12.9	7.8	4.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	4.8	6.6	9.4	10.3	9.8	10.1	10.2	8.8	6.6	4.0	3.9		
		MCDBR	4.0	5.3	10.2	13.7	13.6	12.9	13.5	13.4	12.0	9.1	5.3	4.2		
	5% WB	MCWBR	3.8	4.3	6.6	7.1	6.7	6.4	5.6	5.5	6.2	6.0	4.3	3.9		
		MCWBR	3.7	5.0	9.0	12.4	11.9	11.2	11.5	11.4	10.4	8.0	4.9	4.1		
(40) Clear-Sky Solar Irradiance	taub		0.299	0.332	0.366	0.413	0.400	0.391	0.428	0.422	0.382	0.357	0.341	0.300		
		taud	2.207	2.181	2.220	2.185	2.289	2.338	2.251	2.264	2.341	2.397	2.292	2.244		
	Ebn at Noon		650	746	804	812	848	859	815	797	783	723	596	573		
		Edh at Noon	64	91	109	129	123	118	127	119	97	75	61	52		
(44) All-Sky Solar Radiation	RadAvg		0.66	1.41	2.67	4.08	5.26	5.54	5.16	4.46	3.09	1.59	0.59	0.41		
	RadStd		0.12	0.26	0.42	0.53	0.53	0.46	0.52	0.40	0.39	0.33	0.06	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.62	N/A	N/A	N/A	+125	+41
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	+0.46	+0.61	N/A	N/A	N/A	N/A

Nomenclature: See separate page